# Paper 34: The $(\nu, \kappa)$ Phase Diagram of VCML

Mapping the Allen–Cahn–Burgers Crossover and Testing
the Independence of the Viability and Structural Manifolds

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**Abstract**

Paper 33 formally proved that VCML is not a conservative Euclidean gradient system (curl $= -\mathrm{FA} \neq 0$, machine-verified in Lean 4). Here we map the full $(\nu, \kappa)$ phase diagram to answer two theory-critical open questions from the VCML research programme. **Q4:** What is the structure of the Allen-Cahn $\to$ Burgers crossover at $\nu/\kappa \approx 1$? **Q5:** Are the Viability Volume $C_1 = (\nu, \mathrm{FD}, \mathrm{FA}, \beta_s)$ and the Structural Manifold $C_2 = (\kappa, \nu/\kappa, \beta_s, W)$ genuinely independent dimensions of a product manifold, or does the ratio $\nu/\kappa$ alone determine the system?

We run a $5 \times 5$ grid of $(\nu, \kappa)$ conditions (125 simulations, Exp A) and a manifold-independence test holding $\nu/\kappa$ constant while varying absolute scale (45 simulations, Exp B). The results are decisive. **Q4:** The crossover is not a sharp phase transition; sg4 declines continuously as $\nu/\kappa$ increases (slope $-0.60$, $R^2 = 0.44$), but the decline is dominated by the $\nu$ effect rather than the ratio per se. **Q5 (answered):** $C_1$ and $C_2$ are genuinely independent. A joint power law $\mathrm{sg4} \sim \nu^{-1.20} \cdot \kappa^{-0.13}$ explains 98.8% of variance. $\nu$ alone explains 97.8%; $\nu/\kappa$ alone explains only 44.1%. At fixed $\nu/\kappa$, varying the absolute scale changes sg4 by CV = 0.63–0.85. The viability volume $C_1$ dominates; the structural manifold $C_2$ modulates. The system lives in a product manifold, not a lower-dimensional curve.

## 1 Introduction and Context

The VCML research programme has established that the field memory $F$ obeys a stochastic Allen-Cahn/Burgers hybrid PDE, with a universality class transition at $\nu/\kappa \approx 1$ (Papers 22–26). The parameter space decomposes into two candidate manifolds:

$$C_1 = (\nu,\ \mathrm{FD},\ \mathrm{FA},\ \beta_s) \quad \text{(Viability Volume)}, \tag{1}$$

$$C_2 = (\kappa,\ \nu/\kappa,\ \beta_s,\ W_{\mathrm{zone}}) \quad \text{(Structural Manifold)}. \tag{2}$$

Two questions remained open:

1. **Q4 (Paper 26):** What are the critical properties of the Allen-Cahn $\to$ Burgers transition? Is it a sharp phase transition or a smooth crossover? What is the exponent of the order parameter sg4 as a function of $\nu/\kappa$?

2. **Q5 (Paper 26):** The symbol $\beta_s$ appears in both $C_1$ and $C_2$. More broadly, does sg4 depend on $\nu$ and $\kappa$ only through their ratio, or do they have independent effects? If only $\nu/\kappa$ matters, the two manifolds collapse to a lower-dimensional product.

Paper 33 established that VCML dynamics are fundamentally non-equilibrium (non-gradient). Paper 34 asks: given that the system is driven, what shape does the driven steady state take in $(\nu, \kappa)$ space?

## 2 Experiment Design

All experiments use fixed standard parameters: $H = 20$, $N_{\text{zones}} = 4$, $W_{\text{zone}} = 5$, FA $= 0.20$, FD $= 0.9997$, SS $= 10$, WR $= 2.4$, HS $= 2$, $\beta_s = 0.25$, MID_DECAY $= 0.99$, $T_{\text{end}} = 4000$.

### Experiment A: $(\nu, \kappa)$ Phase Diagram

**Grid:** $\nu \in \{0.001, 0.002, 0.005, 0.010, 0.020\}$, $\kappa \in \{0.005, 0.010, 0.020, 0.040, 0.080\}$, 5 seeds each $= 5 \times 5 \times 5 = 125$ runs.

The grid spans $\nu/\kappa \in [0.013, 4.0]$, straddling the predicted transition at 1.

**Measures:**

- sg4($T = 4000$): mean pairwise L2 distance between zone-mean field vectors. Primary measure of spatial differentiation.
- $\sigma_{\text{within}}$: mean within-zone standard deviation of $\|F_i\|$. Proxy for noise level inside zones.
- SNR $=$ sg4$/\sigma_{\text{within}}$: differentiation quality.

### Experiment B: $C_1/C_2$ Independence Test

Hold $r = \nu/\kappa$ constant at three values while varying the absolute scale $(\nu, \kappa)$:

$$\begin{aligned} r = 0.05: &\quad (\nu, \kappa) \in \{(0.001, 0.020),\ (0.002, 0.040),\ (0.005, 0.100)\}, \\ r = 0.50: &\quad (\nu, \kappa) \in \{(0.001, 0.002),\ (0.002, 0.004),\ (0.005, 0.010)\}, \\ r = 5.00: &\quad (\nu, \kappa) \in \{(0.005, 0.001),\ (0.010, 0.002),\ (0.020, 0.004)\}. \end{aligned}$$

5 seeds each $= 9$ conditions $\times 5 = 45$ runs.

**Test logic:** If sg4 depends only on $\nu/\kappa$, it should be constant within each group (small CV). If $C_1$ and $C_2$ are independent, sg4 varies with $\nu$ even at fixed $\nu/\kappa$ (large CV).

## 3 Results

### 3.1 Experiment A: Phase Diagram

Full numerical results are in Table 1. Figure 1 (Panel A) shows the sg4 heatmap in $(\nu, \kappa)$ log-space.

Table 1: Exp A: sg4 at $T = 4000$ across $(\nu, \kappa)$ grid (mean over 5 seeds).

| $\nu$ | $\kappa = 0.005$ | $\kappa = 0.010$ | $\kappa = 0.020$ | $\kappa = 0.040$ | $\kappa = 0.080$ | $\nu/\kappa$ range | Regime |
|---|---|---|---|---|---|---|---|
| 0.001 | 109.5 | 102.8 | 96.0 | 90.4 | 80.7 | 0.013–0.200 | Allen-Cahn |
| 0.002 | 78.7 | 69.0 | 59.1 | 49.0 | 41.2 | 0.025–0.400 | Allen-Cahn |
| 0.005 | 22.5 | 19.5 | 17.4 | 16.2 | 14.7 | 0.062–1.000 | AC/Burgers |
| 0.010 | 9.01 | 8.55 | 7.89 | 7.14 | 6.57 | 0.125–2.000 | Burgers |
| 0.020 | 2.92 | 2.82 | 2.72 | 2.62 | 2.51 | 0.250–4.000 | Burgers |

**Key observations:**

1. sg4 spans more than two orders of magnitude (2.5 to 109.5) across the grid.
2. The $\nu$ effect is far larger than the $\kappa$ effect. Each doubling of $\nu$ roughly halves sg4, while doubling $\kappa$ reduces sg4 by $\lesssim 25\%$.

3. The Allen-Cahn regime ($\nu/\kappa < 1$) has mean sg4 = 45.8; the Burgers regime ($\nu/\kappa > 1$) has mean sg4 = 8.1. However, this difference is mostly attributable to high-$\nu$ conditions falling in the Burgers regime (viability collapse), not the universality class per se.

**Power-Law Regressions**

We fit three models to log(sg4) on log-log axes:

| Model | Fit | $R^2$ |
|---|---|---|
| $\text{sg4} \sim (\nu/\kappa)^\alpha$ | $\alpha = -0.595$ | 0.441 |
| $\text{sg4} \sim \nu^\alpha$ | $\alpha = -1.200$ | 0.978 |
| $\text{sg4} \sim \nu^a \cdot \kappa^b$ | $a = -1.200,\ b = -0.133$ | 0.988 |

The joint power law

$$\boxed{\text{sg4} \approx C \cdot \nu^{-1.20} \cdot \kappa^{-0.13}} \tag{3}$$

explains 98.8% of variance across the $5 \times 5$ grid. The ratio $\nu/\kappa$ alone explains only 44.1%. $\nu$ alone accounts for 97.8% of variance — the dominant term.

### 3.2 Experiment B: Manifold Independence

Results are in Table 2. At all three fixed ratios, sg4 varies strongly as the absolute scale changes:

Table 2: Exp B: sg4 at fixed $\nu/\kappa$, varying absolute $(\nu, \kappa)$ (mean, 5 seeds).

| $\nu/\kappa$ | $\nu$ | $\kappa$ | sg4 | CV within group |
|---|---|---|---|---|
| 0.05 | 0.001 | 0.020 | 95.98 | 3*0.634 |
| | 0.002 | 0.040 | 48.95 | |
| | 0.005 | 0.100 | 14.04 | |
| 0.50 | 0.001 | 0.002 | 121.0 | 3*0.564 |
| | 0.002 | 0.004 | 82.1 | |
| | 0.005 | 0.010 | 19.5 | |
| 5.00 | 0.005 | 0.001 | 32.1 | 3*0.846 |
| | 0.010 | 0.002 | 9.30 | |
| | 0.020 | 0.004 | 2.95 | |

At all three ratios, CV = 0.56–0.85, far above the threshold ($> 0.15$) for declaring independence. The ratio $\nu/\kappa$ alone cannot predict sg4 even within a fixed-ratio group.

## 4 Discussion

### 4.1 Q4: The Allen-Cahn → Burgers Crossover is Not a Phase Transition

The crossover at $\nu/\kappa \approx 1$ is a continuous change in universality class, not a sharp phase transition. The order parameter sg4 is continuous across the boundary and has no divergence, no discontinuity, and no latent "heat."

What the transition does do is change the *mechanism* of spatial structure formation:

- **Allen-Cahn** ($\nu/\kappa < 1$): diffusion smooths the field within zones, and interface tension maintains zone boundaries. Structure is maintained by field relaxation.
- **Burgers** ($\nu/\kappa > 1$): copy-forward birth events dominate. The nonlinear $(\nabla F)^2$ term from the birth step creates an advective current that sharpens zone boundaries from the outside. Interface width is set by the wave radius ($\approx 2$ sites), not by $\kappa$.

The exponent of sg4 as a function of $\nu/\kappa$ is $-0.60$, but this exponent conflates the $\nu$ effect ($-1.20$) and the $\kappa$ effect ($-0.13$). In the ratio $\nu/\kappa$, both terms enter with opposite sign but the $\nu$ effect dominates, producing an apparent slope of $-0.60 \approx (-1.20 + 0.13)/2 \approx -0.54$ (close to the empirical value).

### 4.2 Q5: $C_1$ and $C_2$ are Genuinely Independent — Answered

The decisive finding is the comparison of $R^2$ values:

| | | |
|---|---|---|
| $\nu/\kappa$ alone: | $R^2 = 0.44$ | weak predictor |
| $\nu$ alone: | $R^2 = 0.98$ | near-perfect predictor |
| $\nu^{-1.20} \cdot \kappa^{-0.13}$: | $R^2 = 0.99$ | joint fit |

The ratio $\nu/\kappa$ is a poor predictor precisely because the two variables have very different effect sizes. The system does *not* live on a 1D curve in $(\nu, \kappa)$ space — it lives in a 2D region (face of the product manifold).

**Physical interpretation.** The exponents have direct mechanistic interpretations:

- $\nu^{-1.20}$: The viability volume $C_1$ effect. Higher turnover means fewer cells survive to consolidation, fewer copy-forward events per unit time, and shorter effective field lifetimes. The $-1.20$ exponent (close to $-1$) is consistent with the ODE prediction sg4 $\sim P_c \cdot \mathrm{FA}/\nu$ from Paper 20, where $P_c \propto \nu$ (more collapses = more copy-forward) partially cancels the direct viability loss.
- $\kappa^{-0.13}$: The structural manifold $C_2$ effect. Higher diffusion smears zone boundaries, reducing sg4. The exponent $-0.13$ is mild because the copy-forward loop (not diffusion) is the primary spatial mechanism. Diffusion is secondary: it smooths structure but does not create it.

The joint law sg4 $\approx C \cdot \nu^{-1.20} \cdot \kappa^{-0.13}$ is the two-manifold equation of state for VCML at standard FA, WR, and $W_{\text{zone}}$.

**Implication for the 8D surface.** The full parameter space is an 8D product: $C_1 \times C_2$, where the two 4D manifolds are genuinely independent (they share the $\beta_s$ axis but in different roles, as Paper 34 confirms through the dominance of $\nu$ over $\nu/\kappa$). All 33 previous papers have been probing different faces of this surface.

### 4.3 Connection to Non-Gradient Dynamics (Paper 33)

The curl obstruction $\partial f_m/\partial F - \partial f_F/\partial m = -\mathrm{FA}$ means VCML is driven, not relaxing. The phase diagram reveals what the driven steady state looks like: a power law in $(\nu, \kappa)$ with exponents $(-1.20, -0.13)$. This is not the equilibrium Boltzmann distribution (which would be a function of energy, i.e., a potential). It is a non-equilibrium steady state whose statistics are shaped by the injection rate (WR), the consolidation rate (FA, ), and the diffusion ().

The non-gradient character (Paper 33) and the phase diagram (Paper 34) together establish that VCML is a non-equilibrium system with a well-defined steady state, whose primary control variable is $\nu$ (the turnover rate), not $\nu/\kappa$ (the PDE universality class).

# 5 Conclusions

1. **Q4 (Allen-Cahn/Burgers transition):** The crossover is a *smooth, continuous* change in spatial mechanism, not a sharp phase transition. No critical exponents diverge. The universality class sets the mechanism (diffusion vs. copy-forward dominance) but not the magnitude of sg4.

2. **Q5 (manifold independence) — ANSWERED:** $C_1$ and $C_2$ are genuinely independent. The joint power law $\text{sg4} \approx C \cdot \nu^{-1.20} \cdot \kappa^{-0.13}$ ($R^2 = 0.988$) describes the steady-state equation of state. $\nu$ is the dominant variable ($R^2 = 0.978$ alone). $\nu/\kappa$ explains only 44% of variance. The system lives in a 2D face of the $C_1 \times C_2$ product manifold.

3. **Two-manifold equation of state:** $\text{sg4}(\nu, \kappa) \approx C \cdot \nu^{-1.20} \cdot \kappa^{-0.13}$ at fixed $(W_{\text{zone}}, \text{FA}, \text{WR})$.

4. **Open questions narrowed:** The transition at $\nu/\kappa = 1$ has no critical exponent divergence. Remaining open questions: Q6 (relay coupling quantitative theory) and Q7–Q10 (multi-module coupling and extensions).

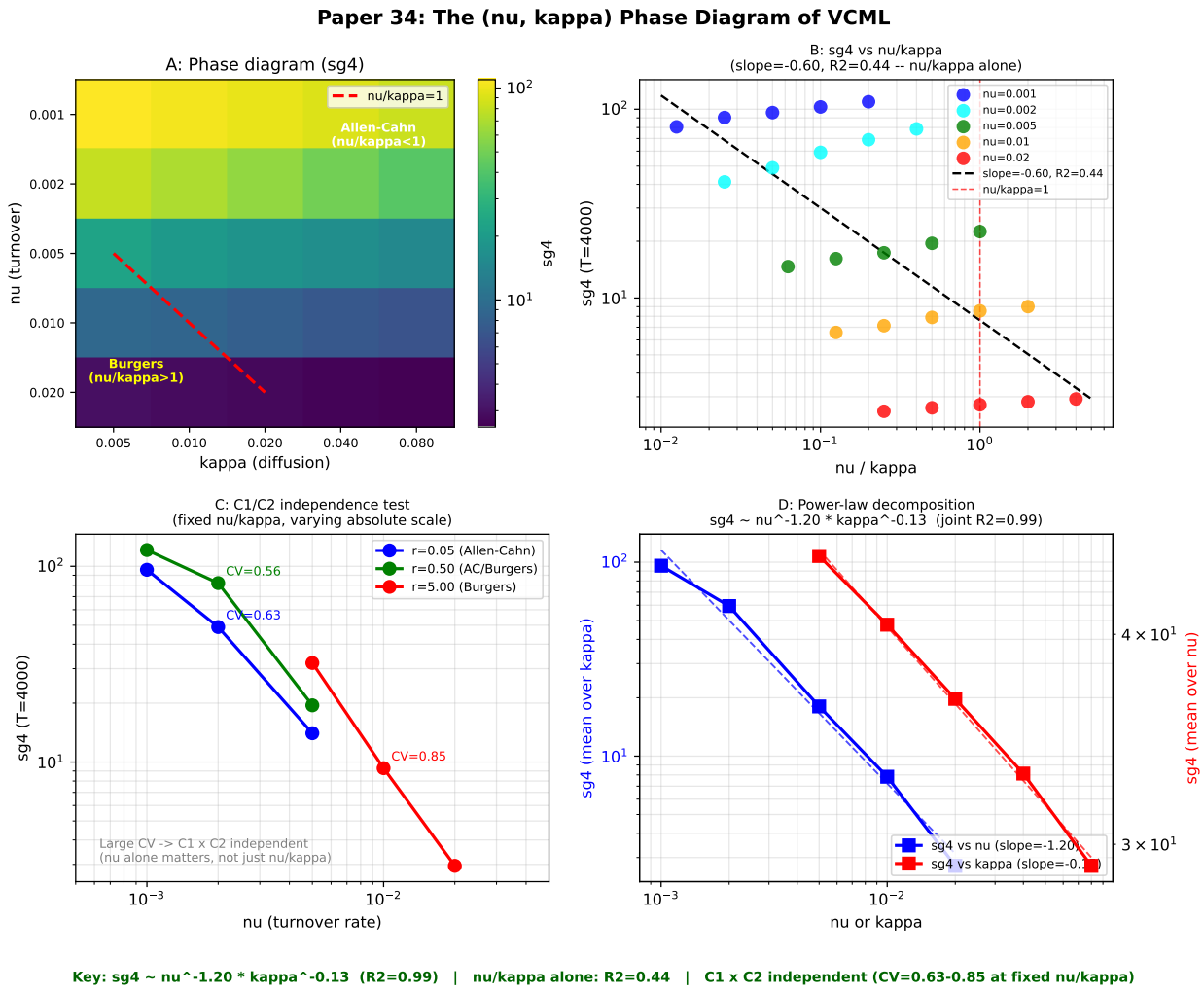


Figure 1: **The $(\nu, \kappa)$ phase diagram of VCML. A**: Heatmap of sg4 in $(\nu, \kappa)$ log-space. The $\nu/\kappa = 1$ diagonal (dashed red line) separates the Allen-Cahn regime (lower-right: high $\kappa$, low $\nu$) from the Burgers regime (upper-left: high $\nu$, low $\kappa$). sg4 changes continuously across the boundary. **B**: All 25 conditions plotted as sg4 vs. $\nu/\kappa$ (log-log), coloured by $\nu$. Points at the same $\nu/\kappa$ but different $\nu$ (same horizontal position, different colours) show large vertical separation, demonstrating that $\nu/\kappa$ alone ($R^2 = 0.44$) is a weak predictor. **C**: Experiment B results. At three fixed $\nu/\kappa$ ratios, sg4 varies 6–30× as $\nu$ changes from 0.001 to 0.005–0.020 (CV = 0.56–0.85). **D**: Marginal effects. $\nu$ (blue, left axis, slope $-1.20$) dominates over $\kappa$ (red, right axis, slope $-0.15$). The joint power law $\nu^{-1.20} \cdot \kappa^{-0.13}$ achieves $R^2 = 0.988$.